
Maintain Integration

Given an integration maintenance use case, identify performance monitoring needs for integration requirements.

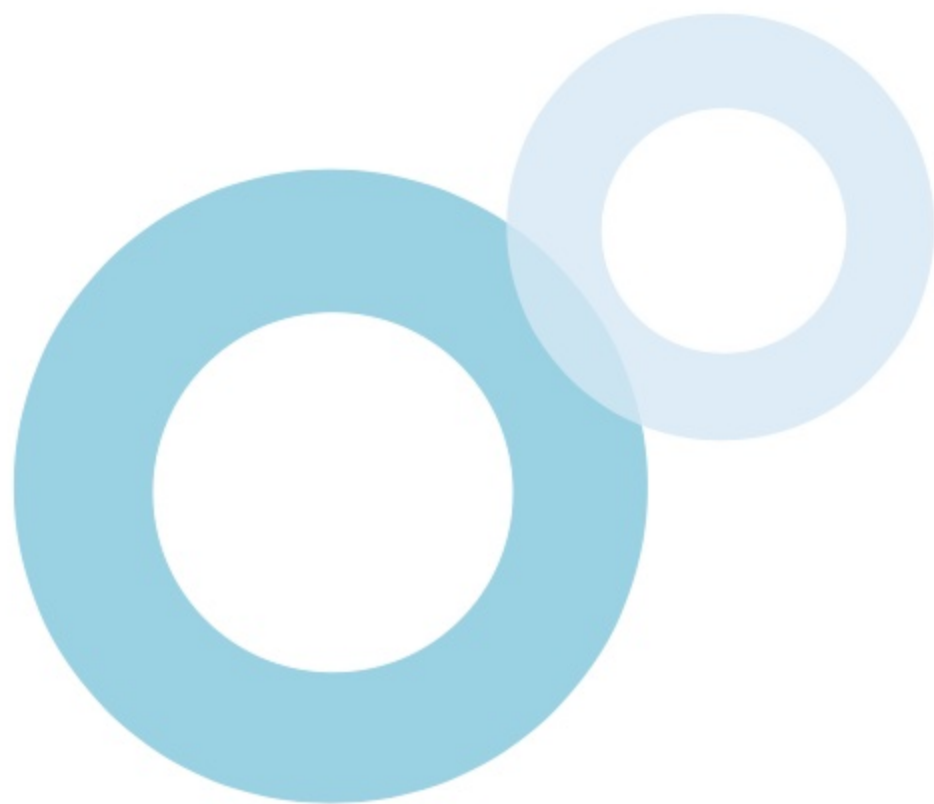
Table of Contents

-  [Event Monitoring](#)
-  [Real-Time Events](#)
-  [Lightning URI Event](#)
-  [Telemetry Timing Events](#)
-  [Experienced Page Time \(EPT\)](#)
-  [Use Cases](#)



After studying this topic, you should be able to:

-  Identify what Event Monitoring is and how it can be used to meet performance monitoring needs
-  Describe what real-time events are, how to configure them, and their capabilities
-  Describe what Experienced Page Time is and how it is used to measure performance



Introduction

Salesforce provides tools that can be used to **evaluate performance** of the org, its applications, or pages. These tools can be used to **monitor** specific attributes and metrics related to integration solutions such as **integration user logins**, **API requests**, **outdated API versions**, and others. In this topic, tools such as **Event Monitoring**, and **real-time events** will be covered to identify their capabilities and determine how they can be used to meet performance monitoring needs in integrated solutions.



Overview

Event Monitoring

Capture granular details of user activities, or events, stored in event log files



Real-Time Events

Stream or store captured events to monitor user activity near real-time



Experienced Page Time

Determine performance by measuring page load time using various methods



Event Monitoring

Table of Contents



Event Monitoring

Event Monitoring, which is included in **Salesforce Shield**, enables tracking of **user activities**, or **events**, in the org. The tool provides **granular details** of activity data that can be used for:

❖ DATA INTELLIGENCE

User logins, accessed reports, loaded pages, searches performed, page interactions, callouts executed, and more can be analyzed to gain insights.

❖ DATA SECURITY

Event trends can be monitored in order to identify any abnormal behavior or suspicious user activities to protect data.

❖ OPTIMIZATION

User traffic and workload can be analyzed and reported errors investigated in order to improve and optimize org performance.



Tracked Events

There are over **50 types** of events that can be tracked using the **Event Monitoring** tool. Some of these tracked events are grouped or described below.



LOGIN / LOGOUT

Login and logout details of each user in the org



URI

User interaction with the web browser UI



PERFORMANCE

Lightning Experience and Mobile App performance



ERRORS

Lightning Experience and Mobile App errors



PAGE LOADS

Accessing or loading of Visualforce pages



API CALLS

SOAP, REST, and BULK API calls and total usage



APEX EXECUTIONS

Executions, callouts, triggers, exceptions, etc.



REPORTS

Loading and exporting of reports

Considerations

The following describes some considerations when using the **Event Monitoring** tool.

❖ API-ONLY FEATURE

The **Event Monitoring** tool is an API-only feature and access to its data is not available in the Salesforce user interface, except through the **Event Monitoring Analytics** app, which is a built-in feature in Salesforce.

❖ EVENT LOG FILES

Each captured event generates an event log file. Event log files are stored in an API standard object called **EventLogFile**. The log data is contained in the field of the object called **LogFile** and is stored in *base64* format.



Considerations

The following describes some considerations when using the **Event Monitoring** tool.

❖ EVENT TYPES

Each EventLogFile record is associated with an **event type**. The fields or structure of the data contained in the log file varies based on the event type. The field type and names are specified in the **LogFileFieldTypes** and **LogFileFieldNames** fields of the EventLogFile record respectively.

❖ EVENT TYPE ACCESS

Not all access to event types are free. While Developer Edition orgs have free access to **all log types**, the Enterprise, Unlimited, and Performance Edition orgs have access only to **insecure external assets, login/logout data**, and **total API usage logs** by default. Access to the event log files of the other event types requires an extra cost for these orgs.



Considerations

The following describes some considerations when using the **Event Monitoring** tool.

❖ LOG GENERATION

An event log file is generated and available only **24 hours after** the user activity, or event, has occurred.

❖ LOG RETENTION

Event log files are automatically deleted **after 1 day**. For an extra cost, The retention period can be extended to **30 days** for Enterprise, Unlimited, and Performance Edition orgs.



How to Access Event Log Files

Aside from Event Monitoring Analytics app, Salesforce provides other tools for **accessing** event log files.



ELF BROWSER

The **Event Log File (ELF) Browser** in Setup can be used to easily **view** and **download** event log files. To download, a date range is specified and the **Download as CSV File** option is selected. Up to **1 year** of log data is available for download.



DEVELOPER CONSOLE

To view Event Logs in Developer Console, go to *File > Open > Objects*. Select **EventLogFile** under **Entities** and click *Open*. Select the necessary fields on the object, click on the **Query** button to generate the SOQL statement, and click on **Execute**.



REST/SOAP API

To retrieve event monitoring data using the REST API, the **Query** resource is used. For example: **/services/data/v55.0/query?q=[soql]** where [soql] represents a SOQL query on the **EventLogFile** object. For SOAP API, the **query()** call is used.

How to Visualize Event Log Files

Visual representation of data helps in **identifying patterns** and **analyzing user behavior**. The event log files are typically downloaded, and then uploaded and viewed in a **data visualization tool**.



DOWNLOAD

Event log files are available for download in CSV format using the **ELF browser** in Setup. Alternatively, the file can be downloaded directly via the URL:

<DOMAIN>/servlet/servlet.FileDownload?file=<ID>
where <DOMAIN> is the org domain and <ID> is the record ID of the event log file.



VISUALIZE

The data can be visualized using the **Event Monitoring Analytics app**. This option automatically loads the Event Log Files into the app, so a download is **not required**. Other tools that can be used, for example, are *Splunk App*, *FairWarning*, *New Relic Insights*, and other third-party or custom data visualization apps.

Real-Time Events

[Table of Contents](#) 

Real-Time Events

Real-Time Events is a feature in **Event Monitoring** which enables tracking user activities of certain event types in **near real-time**. The following event type **categories** support real-time events.

❖ AUTHENTICATION

Real-time events are fired, for example, when:

- A user logs in from an unusual **browser** or **IP address**
- A user logs out of Salesforce by clicking on the **Log Out** button
- A Salesforce administrator logs in as **another user**

❖ DATA ACCESS

Real-time events are fired, for example, when:

- Accessing, updating, or exporting a **list view**
- Creating, running, updating, or exporting a **report**
- Querying **sensitive** data
- Downloading the result of a **Bulk API request**



Real-Time Events

Real-Time Events is a feature in **Event Monitoring** which enables tracking user activities of certain event types in **near real-time**. The following event type **categories** support real-time events.

❖ THREAT DETECTION

Real-time events are fired when a security vulnerability is detected:

- **Credential stuffing** - successful user login due to large-scale automated login requests using stolen data
- **Report anomalies** - when a user unusually accesses or exports reports based on the activity history of the user
- **Session hijacking** - when unauthorized users take over a session after successfully stealing a valid session identifier



Real-Time Events

Real-Time Events is a feature in **Event Monitoring** which enables tracking user activities of certain event types in **near real-time**. The following event type **categories** support real-time events.

❖ PAGE ACCESS

A real-time event can be generated **every time** a user **accesses a page**.

❖ MOBILE SECURITY EVENTS

Real-time events can be fired, for example, when a **Salesforce mobile app user** takes a screenshot. These mobile app-related events are part of **Enhanced Mobile Security** and requires additional licensing.



Configuring Real-Time Events

Real-time events can be enabled in the **Event Manager** in Setup. Once a real-time event is **enabled**, there are **3 ways** how to handle the event each time it is fired.



STREAMING

A real-time event can be published as a **streaming event**. For example, an external monitoring application can subscribe to receive an **event message** whenever a user logs in from a new location.



STORING

A real-time event can be stored in a **big object** table and available in the org for 6 months. The data can then be accessed using **Bulk API**, **Batch Apex**, or regular **SOQL** if the primary index and event date is available.



SECURITY

A real-time event can be evaluated against a **transaction security policy** related to API, list view, login, and report events. If the policy's **criteria is met**, a login request can be rejected, for example.

Event Manager

For example, a captured **API Anomaly Event** can be **stored** as a big object which can be viewed in a report, and **streamed** to notify a streaming client in near real-time.

Event Manager

Events

Event Manager

Didn't find what you're looking for?
Try using Global Search.

SETUP

Event Manager

Events

17 events • Sorted by Name

Name	Subscription Channel	Type	Description	Streaming Da...	Storing Data
API Anomaly Event	/event/ApiAnomalyEvent	Event Monitoring	Track anomalies in how users make API ...		
API Event	/event/ApiEventStream	Event Monitoring	Track user API queries in your org		
Bulk API Result Event	/event/BulkApiResultEvent	Event Monitoring	Track when a user downloads the result...		
Concurrent Long Running Apex...	/event/ConcurLongRunApe...	Event Monitoring	Track when a Concurrent Long Running ...		Not applicable
Credential Stuffing Event	/event/CredentialStuffingE...	Event Monitoring	Track when a user successfully logs in t...		
Identity Provider Event	Not applicable	Event Monitoring	Track identity provider activities	Not applicable	
Identity Verification Event	Not applicable	Event Monitoring	Track when users verify their identity	Not applicable	
Lightning URI Event	/event/LightningUriEventSt...	Event Monitoring	Track when a user creates, accesses, u...		
List View Event	/event/ListViewEventStream	Event Monitoring	Track when a user accesses data with li...		

Enable Storage

Enable Streaming

Event Studio

The Event Studio in Setup enables **monitoring** of **event publishing** and **consumption activity** on the Salesforce Event Bus, showing which **publishers** send events and which **subscribers** receive them.

Event Publishers Event Subscribers

Time Range
Last 7 Days Submit

3 items Search this list...

Publish Method	Events Published	Last Publish Time
Flow	298	11/19/2025, 6:36:02 AM
Apex	83	11/19/2025, 4:55:56 AM
System	9,009	11/19/2025, 10:09:13 AM

The available **Time Range** filter options are the **last 15 minutes**, **last hour**, **last 24 hours**, and **last 7 days**.

Event Publishers **Event Subscribers**

Time Range
Last 7 Days Submit

2 items Search this list...

Subscription Me...	Number Of Clients	Events Delivered	Last Delivery Time
CometD	215	2,578	11/19/2025, 4:28:40 AM
Apex Trigger	3	312	11/19/2025, 6:36:09 AM

Lightning URI Event

[Table of Contents](#)



Lightning URI Event

When a user **creates**, **accesses**, **updates**, or **deletes** a record in Lightning Experience, the activity is detected and a **real-time event** can be created or published.



STORED OR STREAMED

When detected, the event can be created and stored as a big object record called **LightningUri**, or it can be published as a streaming event called **LightningUriEventStream**.



PAGE LOAD TIMES

Lightning URI events can be used to **monitor** and **analyze** page load times in the org and determine what causes changes in the **performance** of a **Lightning page**.

Lightning URI Event Fields

The following describes a few of the Lightning URI event fields that can be used for tracking **page performance**. Certain factors that affect the load time of a page is called a **deviation**.



EffectivePageTime

Time (ms) taken to **complete loading** of a page before a user can **fully interact** with the page



HasEffectivePageTimeDeviation

Returns true when a **deviation** has been **detected**; otherwise, false is returned



EffectivePageTimeDeviationReason

Indicates the **reason** for the **page deviation** that affected the load time



EffectivePageTimeDeviationErrorType

When the deviation is caused by an **error**, this field indicates the **origin** of the error

Page Deviation Reasons

A deviation impacts the normal loading process of a page. The following are deviation types or factors that **cause a deviation**, which are indicated in the **EffectivePageTimeDeviationReason** field.



PageInDom

When the page was loaded from the **browser cache**



PageHasError

When an undefined **page loading error** was encountered



PageNotLoaded

When the user **navigates away** before the page load completes



PreviousPageNotLoaded

Navigating to a **new page** before the **previous page** completely loads causes deviation on the new page



InteractionsBeforePageLoaded

When the **user interacts** with the page **before** it has completely loaded



PageInBackgroundBeforeLoaded

When a **background loading process** was run on a page.

Telemetry Timing Events

Table of Contents



UI Telemetry Timing Events

Detailed network performance metrics can be captured using the **Resource Timing** and **Navigation Timing** event log file types.



RESOURCE TIMING

The Resource Timing event log file type is used to **measure the duration** required by a **browser** to retrieve **resources** such as files, images, etc. from a **remote server**.



NAVIGATION TIMING

The Navigation Timing event log file type is used to **monitor metrics** associated with **page navigation**, including the duration a browser requires to build a page's **Document Object Model (DOM)**.

UI Telemetry Resource Timing Event

The following lists only a few of the **fields and data** that is captured in a resource timing event, which is triggered when **loading an application resource** from a server.



APP_NAME

Name of the **application** that the user accessed and **initiated** the request.



BROWSER_NAME

Name of the **browser** that the user used **during** the request.



CONNECTION_TYPE

The **type of network** connection used **during** the request.



DURATION

The **total time** taken to **start** and **complete** loading the resource.



OS_NAME

Name of the **operating system** of the current user's **device**.



RESPONSE_STATUS

The HTTP response status code used to indicate the **state of the resource**.



TRANSFER_SIZE

The computed size of the the **resource** and **request** & **response headers**.



URL

The URL of the request used to **access the resource**.

UI Telemetry Navigation Timing Event

The following lists only a few of the **fields and data** that is stored in a navigation timing event, which captures performance metrics related to **navigation** such as a page's **DOM generation**.



APP_NAME

Name of the **application** that the user accessed and **initiated** the request.



BROWSER_NAME

Name of the **browser** that the user used **during** the request.



DOM_INTERACTIVE

Time taken to **load the page** whether or not resources finished loading.



DOM_COMPLETE

Time taken to **completely load** the **page** and all its **resources**.



LOAD_EVENT_START

Time taken for the page's **load event handler** to **start**.



LOAD_EVENT_END

Time taken for the page's **load event handler** to **complete**.



NAVIGATION_TYPE

The type of navigation such as browser **reload**, **back/forward**, etc.



REDIRECT_COUNT

Captures the **number of redirects** taken before the final navigation.

Experienced Page Time (EPT)

[Table of Contents](#)



Experienced Page Time (EPT)

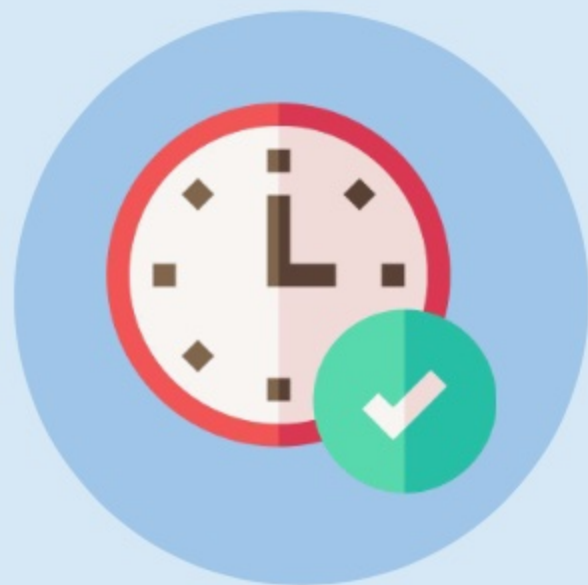
Page performance is measured in **EPT**, which is the **page load time** that users experience before the loaded page is ready for use. Salesforce provides **4 ways** to measure EPT.

❖ EPT COUNTER

An EPT counter can be displayed on the **header section** in Salesforce to quickly determine the load time (in seconds) of the current page. To achieve this, either methods can be used.

- Append the **?eptVisible=1** parameter to the URL of the current page.
- Enable the **Lightning Component Debug Mode** in Setup.

Enabling the Lightning Component Debug Mode will affect page performance since caching **will be disabled** and also, component code **will not be minified**, although it will display network bandwidth consumption in addition to the EPT count. Using the *eptVisible* parameter to display the EPT count has far much lesser impact on performance.



Experienced Page Time (EPT)

Page performance is measured in **EPT**, which is the **page load time** that users experience before the loaded page is ready for use. Salesforce provides **4 ways** to measure EPT.

❖ LIGHTNING USAGE APP

The Lightning Usage App, which can be accessed through the **App Launcher**, can be used to determine EPT of pages in the org. The tool, for example, provides an **aggregated report** of the most viewed pages or slowest record pages and their load times. Aside from measuring the EPT according to **pages**, the tool can also measure EPT based on the **browser** that was used to access the pages.



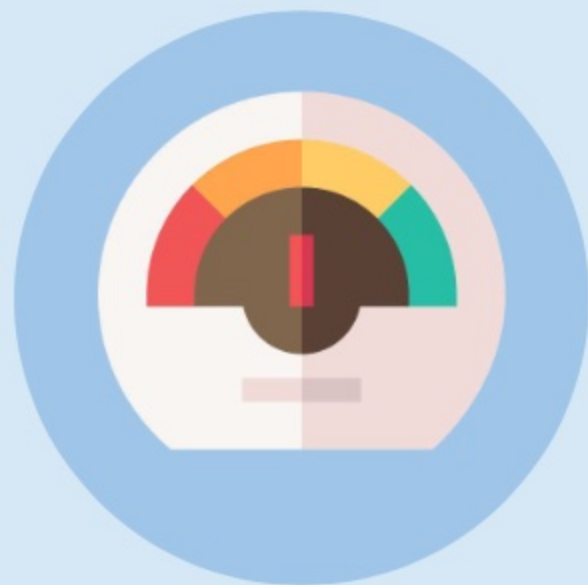
Experienced Page Time (EPT)

Page performance is measured in **EPT**, which is the **page load time** that users experience before the loaded page is ready for use. Salesforce provides **4 ways** to measure EPT.

❖ CUSTOM REPORT

Salesforce allows building a custom report using **Lightning Usage App** objects in order to analyze page performance. For example, a custom report type can be created using the **LightningUsageByBrowserMetrics** object to evaluate how the most viewed pages perform according to the used browser. Other Lightning Usage App objects that can be used are:

- **LightningUsageByAppTypeMetrics** - number of users on Lightning Experience and Salesforce Mobile
- **LightningUsageByPageMetrics** - most viewed standard pages
- **LightningUsageByFlexiPageMetrics** - most viewed custom pages



Experienced Page Time (EPT)

Page performance is measured in **EPT**, which is the **page load time** that users experience before the loaded page is ready for use. Salesforce provides **4 ways** to measure EPT.

◆ EVENT MONITORING

The **Event Monitoring tool** can track user activities in Salesforce related to page access. For example, the browser used and page load time when a user visited a page is captured in a **Lightning Page View** event type. Other event types that can be used for measuring performance are:

- **Apex REST API** - tracks every Apex REST API request
- **Lightning Error** - tracks user errors in Lightning Experience pages and Salesforce mobile app
- **Lightning Interaction** - tracks user actions in Lightning Experience pages and the Salesforce mobile app
- **Lightning Performance** - tracks event trends in Lightning Experience pages and the Salesforce mobile app



Use Cases

Table of Contents 

Scenario



SCENARIO

An integration solution has been developed that implements a remote call-in pattern to enable an external application to query and retrieve products information from Salesforce using the REST API. The queried data will be used to build a product catalog portal on the external application. The company would like to ensure that data in their Salesforce org are secure and that any performance issues are reported by monitoring activities and identifying traffic patterns or peak usage in the integration.



Solution



SOLUTION

Event Monitoring, which is part of Salesforce Shield, can be used to monitor activities, detect event trends, and identify malicious or unusual events that allow the company to take necessary actions. The tool, for example, is capable of capturing every **REST-specific request** from the external application based on the **User Id** of the integration user. Login events and other activities performed by the user can be captured. The collected **Event Log File** data can then be analyzed in a data visualization tool to easily spot activity spikes or unusual patterns based on activity history.



Scenario



SCENARIO

A data breach has occurred where thousands of records containing sensitive data such as login credentials, API keys, etc. have been stolen. A company has determined that one of their integration clients may potentially be a victim in the incident. Before taking other precautionary steps, the company would like to be notified whenever there are unusual login activities coming from the client application that is integrated with their Salesforce org to protect its data and performance.



Solution



SOLUTION

Real-Time Events, which is an added capability of Event Monitoring, can be enabled in Salesforce. In this scenario, real-time events can be configured to fire platform events to notify the administrator, for example, through an email from a platform event-triggered flow, whenever the integration user logs into Salesforce from an unusual **browser** or **IP address**. A platform event can also be fired when **credential stuffing** is detected in the org. Credential stuffing is a cyberattack method using automated logins based on compromised login credential data in order to break into a system.



Learn More

-  [Event Monitoring](#)
-  [Introduction to Real-Time Event Monitoring](#)
-  [Real-Time Event Monitoring](#)
-  [EventLogFile](#)
-  [EventLogFile Supported Event Types](#)
-  [Measure Performance for Your Salesforce Org](#)